AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MT132 – M132
Linear Algebra

Midterm Examination (MTA)- Makeup
Spring Semester 2024/2025

Date: 23 /April/2025

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed: 90 Minutes**

Instructions:

- Total Marks: **30**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

AOU EGYPT

CamScanner

Q-1: [4×2 marks] Choose the correct answer:

1.1: If $A = \begin{bmatrix} 3 & 0 \\ 2 & -1 \end{bmatrix}$, then $2A(A - I) =$

a) $\begin{bmatrix} 12 & 0 \\ 4 & 0 \end{bmatrix}$
b) $\begin{bmatrix} 6 & 0 \\ 2 & 2 \end{bmatrix}$
c) $\begin{bmatrix} 12 & 0 \\ 4 & 4 \end{bmatrix}$
d) $\begin{bmatrix} -12 & 0 \\ -4 & -4 \end{bmatrix}$
e) None of the above

1.2: If A, B and C are square matrices of order n, then $[A^{-1}B + C^{-1}]^{-1} =$

a) $B^{-1}A + C$
b) $B^{-1}A^{-1} + C$
c) $C^{-1} + B^{-1}A$
d) $C + B^{-1}A$
e) None of the above

1.3: If $X = \begin{bmatrix} 2 \\ 0 \\ -3 \end{bmatrix}, Y = \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix}$ and $Z = \begin{bmatrix} -1 \\ 3 \\ 0 \end{bmatrix}$, such that $X = aY - bZ$, then

a) $a = -3, b = -2$
b) $a = 3, b = 2$
c) $a = 3, b = -2$
d) $a = -3, b = 2$
e) None of the above

1.4: If $A = \begin{bmatrix} 3k & 6 \\ 6 & k-1 \end{bmatrix}$ is non-singular, then

a) $k = -4, 3$
b) $k = 4, -3$
c) $k \neq -4, 3$
d) $k \in Z^+$
e) None of the above

AOU EGYPT

CamScanner

Q–2: [8 marks] Answer the following questions as True or False (1 mark), Justifying your answer (3 marks):

2.1: If $\left[\begin{array}{cc|c} a & 3 & -6 \\ 2 & 6 & -12 \end{array}\right]$ is the augmented matrix for a linear system, then the system has infinitely many solutions when $a = 1$.

2.1: If A and B are invertible square matrices then, $(A+B)^{-1} = A^{-1} + B^{-1}$.

Each question is worth 7 marks. Answer the following questions:

Q–3[4+3 marks]:

If $A^{-1} = \begin{bmatrix} 1 & 0 & 1 \\ 2 & -1 & 2 \\ 2 & 2 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 \\ -3 \\ 0 \end{bmatrix}$.

a) Find the matrix A.

b) Solve the linear system $AX = B$.

Q–4[4+3 marks]:

a) Let $A = \begin{bmatrix} 1 & 4 & 1 & 1 \\ 0 & 0 & 0 & 5 \\ 1 & 4 & 2 & -1 \\ 0 & -2 & 0 & 4 \end{bmatrix}$,

i- Is A invertable. Justify your answer.

ii- Find $|2A^4A^{-1}|$.

b) Find a matrix B such that $(B + A^T)A^{-1} = 3I$, where $A = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}$.

End of questions

AOU EGYPT

CamScanner